

TEST YOUR KNOWLEDGE

- 1 Is electromagnetic induction in a circuit necessarily due to a moving **magnet**?
- 2 Does an induced current traverse an open circuit?
- 3 Which physical quantity does a voltmeter measure when connected across an open induction circuit?
- 4 True or false. Correct the false statements.
 - a) The electromagnetic induction phenomenon appears only if there is a relative motion between the induction circuit and the inducing source.
 - b) The induced current lasts as long as the magnetic field varies.
 - c) An inducing source and an induction circuit have to be electrically connected together in order to produce electromagnetic induction.
 - d) Electromagnetic induction takes place only in a variable magnetic field.
 - e) The electromagnetic induction phenomenon is always accompanied by the appearance of an induced e. m. f.
 - f) The electromagnetic induction phenomenon is always accompanied by the appearance of an induced current.

APPLY YOUR KNOWLEDGE

- 5 Choose the best answer:
If the magnetic flux through a circuit increases linearly with time, then the absolute value of the induced em.f
 - a) increases, because the flux is increasing.
 - b) decreases because, according to Lenz's Law, it opposes the original increase of the flux.
 - c) stays the same at all times.
- 6 The plane of a loop of surface area S is held parallel to the lines of a uniform magnetic field $\vec{B}$. The terminals C and D of the loop are connected to a millivoltmeter (Figure 8.20).
 - a) What is the value of the magnetic flux which traverses the loop?
 - b) What is the reading of the millivoltmeter when the loop is moved parallel to the lines of $\vec{B}$?
- 7 The two coils represented in figure 8.21 are not electrically connected together. B_1 is the **source** of induction and B_2 is the induction circuit. Determine the direction of the induced current I_2 when:
 - a) B_2 is moved away from B_1 .

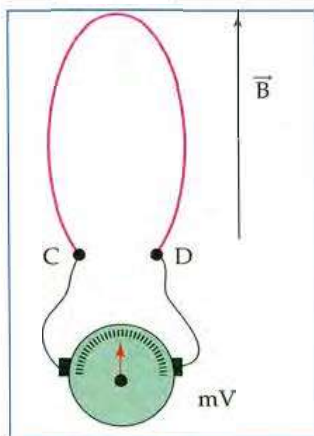


Figure 8.20

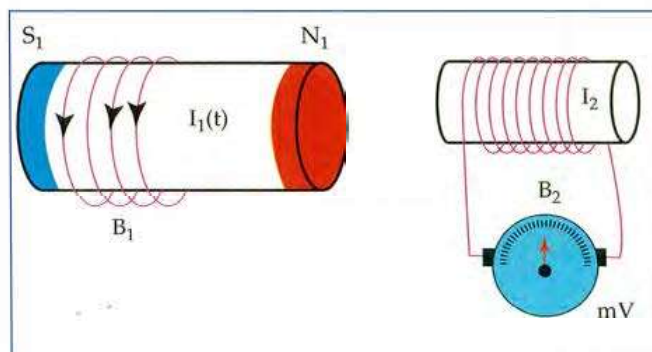


Figure 8.21

- b) B_1 is pushed towards B_2 .
- c) I_1 is decreased.
- d) I_1 is increased.

8 Use Lenz's Law to specify the direction of the induced current in the loop and in the moving conductor in each of the cases shown in figure 8.22.

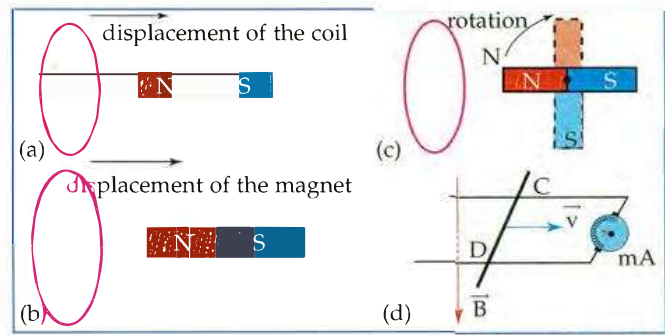


Figure 8.22

PROBLEMS

9 A loop of surface area $s_1 = 10 \text{ cm}^2$, placed in a uniform magnetic field $\vec{B}_1$ of magnitude $B_1 = 8 \times 10^{-2} \text{ T}$, has its plane perpendicular to the field lines. A second loop of area $s_2 = 15 \text{ cm}^2$ is placed in another uniform field $\vec{B}_2$ of magnitude $B_2 = 0.1 \text{ T}$ such that its normal unit vector $\vec{n}$ makes an angle $\theta = 30^\circ$ with the lines of field $\vec{B}_2$.

- 1) Which loop receives more magnetic flux?
- 2) Are there any induced currents in these two loops? Justify.

10 A square loop of side $a = 10 \text{ cm}$ and resistance $R = 0.1 \Omega$ is placed in a uniform magnetic field $\vec{B}$ whose magnitude varies with time as shown in figure 8.23.

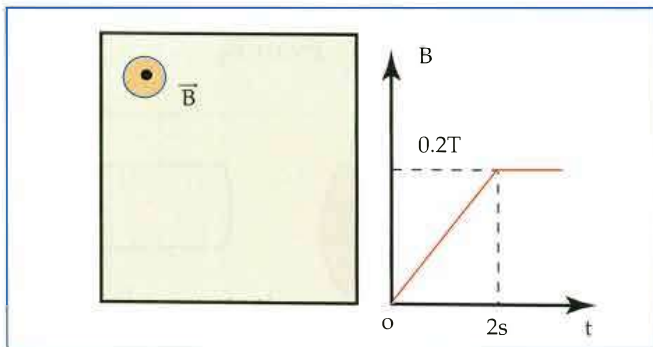


Figure 8.23

- a) Determine, by applying Lenz's Law, the

direction of the induced current for $0 < t < 2\text{s}$, and for $t > 2\text{ sec}$.

- b) After choosing a positive direction along the loop, write down the expression of B as a function of time and calculate, the values of the magnetic flux crossing the loop for $0 < t < 2\text{s}$ and for $t > 2\text{s}$.
- c) Determine the induced e. m. f, the direction and the magnitude of the induced current. Compare the direction of the induced current to that found in part "a" above.
- d) Verify that the direction of the induced current does not depend on the arbitrary choice of the positive direction.

11 A rectangular loop of area S is placed in uniform magnetic field $\vec{B}$ whose magnitude B varies with time. We choose the positive direction as in figure 8.24. The two terminals A and C of the loop are connected to one channel of an oscilloscope.

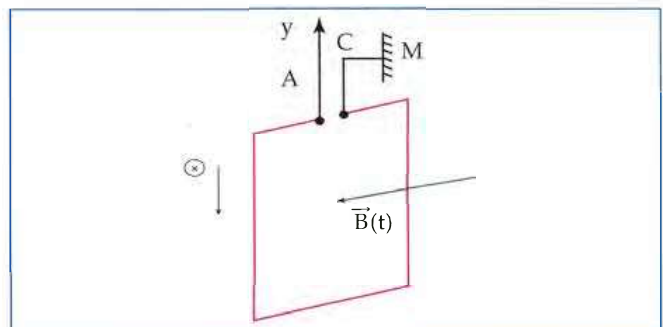


Figure 8.24

- Give the expression for the magnetic flux through the loop.
- Give the expression for the induced e. m. f.
- Does an induced current traverse the loop? Why?
- In which direction is the spot of the oscilloscope shifted when B increases with time?

12 Two horizontal rails are placed at a distance $d = 5 \text{ cm}$ apart. A uniform magnetic field of magnitude $B = 0.4 \text{ T}$ is perpendicular to the plane of the rails. A rectilinear conductor MN slides along the rails, without friction, and with a constant velocity of magnitude $v = 2 \text{ m/s}$ (Figure 8.25). A resistor of resistance $R = 2 \Omega$ is connected to the terminals C and E of the rails. We assume that the resistance of the rails and that of the conductor MN are negligible.

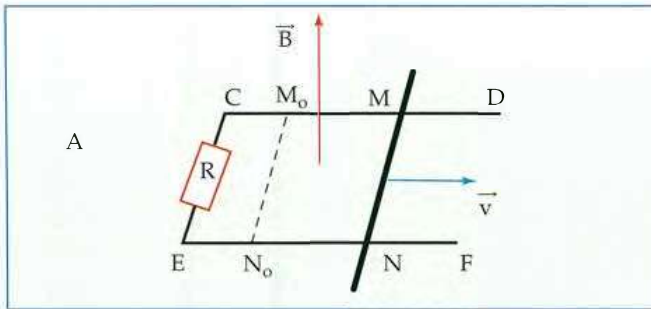


Figure 8.25

- Explain the existence of an induced current in the circuit, and indicate its direction.
- The circuit is oriented in the direction of the induced current. Prove that the magnetic flux through the circuit is expressed by:
 $\phi = \phi_0 + at$, where a is constant.
- Deduce the induced e. m. f. and the induced current in the circuit.
- Determine the characteristics of the

electromagnetic force (Laplace's force) which acts on the moving conductor MN . Compare the mechanical power supplied by the operator (in order to move the rod) to the total electric power generated in the circuit. Conclusion.

13 Consider a solenoid of length $L = 50 \text{ cm}$ and 1500 turns.

- Determine the current for which the magnetic field produced at the center of solenoid has a magnitude of $B = 2 \times 10^{-2} \text{ T}$. ($\mu_0 = 4\pi \times 10^{-7} \text{ SI}$)
- We place a flat circular coil "b" of 250 turns each of area $S = 10 \text{ cm}^2$ inside the solenoid (Figure 8.26)

a - How should the coil be placed in order to receive a maximum magnetic flux? Determine this flux.

b - The coil is now rotated inside the solenoid about a vertical axis with angular velocity of 3000 turns/min. Give the expression of the induced e.m. f within the coil, assuming that the coil receives a maximum flux at $t = 0$.

- The coil is held fixed at its initial position and forms a part of a closed electric circuit. The switch K is opened abruptly. Explain why an induced current traverses the coil. Specify its direction.

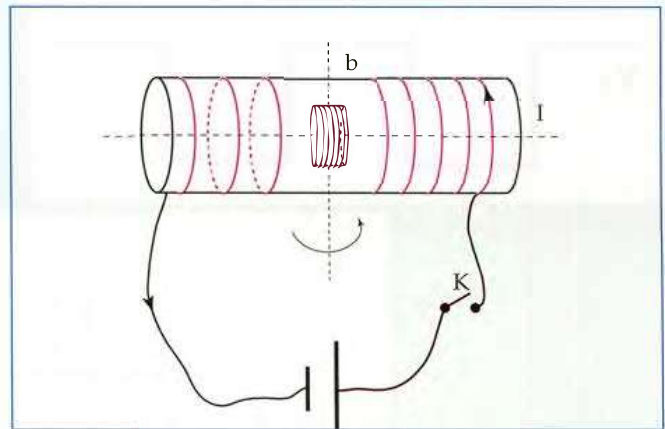


Figure 8.26